TE Ltd. – Project A vs Project B: Resolving the NPV–IRR Conflict

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Case Data and Supporting Calculations

Particulars	Project A	Project B	Basis
Initial investment / project life	₹10,00,000 / 5 years	₹10,00,000 / 5 years	Given
NPV at 12% required return	₹1,86,537	₹1,79,926	Given
IRR	18.52%	20.76%	Given
Present value of cash inflows (NPV + investment)	₹11,86,537	₹11,79,926	Calculated
Profitability index (PV of inflows / investment)	1.1865	1.1799	Calculated
IRR margin above the 12% required return	6.52 percentage points	8.76 percentage points	Calculated
NPV lost per 1 percentage point rise in the rate (approx.)	₹1,86,537 / 6.52 = ₹28,610	₹1,79,926 / 8.76 = ₹20,539	Calculated
Ranking: NPV / IRR	1st / 2nd	2nd / 1st	Conflict

Crossover (Fisher) rate, the discount rate at which both projects have the same NPV, estimated from the two known points of each NPV profile: 12% + ₹6,611 / (₹28,610 - ₹20,539) = **about 12.8%**. Real NPV profiles are slightly curved, so this is an estimate. However, the profiles must cross between 12% and 18.52%, because A is ahead at 12% and B is still positive when A's NPV reaches zero.

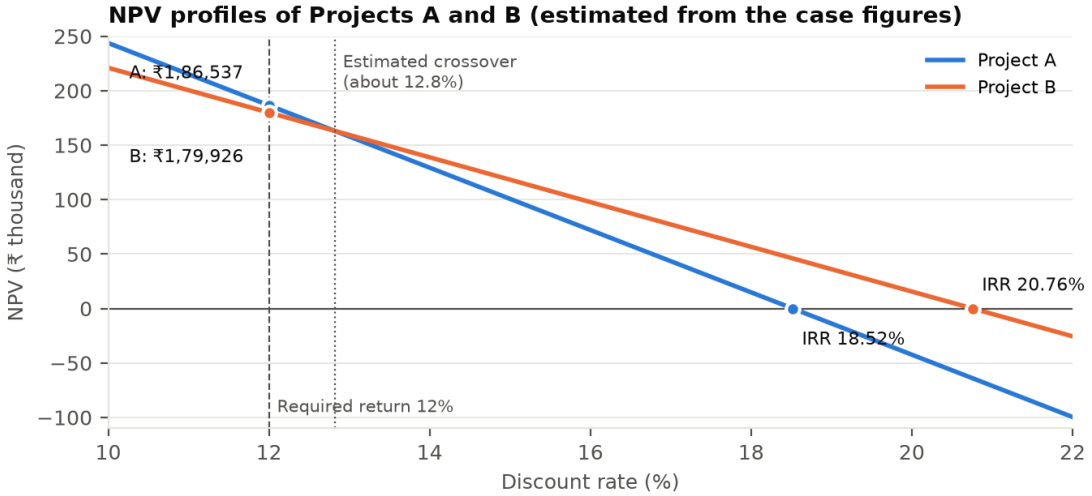


Figure 1 – Estimated NPV profiles. At TE Ltd.'s 12% required return (left of the crossover) Project A has the higher NPV. IRR compares the points where each line crosses zero, far from the 12% rate that actually applies to the company.

Q1. Interpret the Results

A project is acceptable if its **NPV is positive** (the present value of its inflows exceeds the cost) and its **IRR is above the required return**. For a project with normal cash flows (one outflow followed by inflows), the two rules always give the same accept-or-reject answer. Both projects pass both tests. After recovering ₹10,00,000 and earning 12%, A adds ₹1,86,537 and B adds ₹1,79,926 of value in today's money. Their returns (18.52% and 20.76%) exceed the cost of capital, and both profitability indices are above 1 (each rupee invested returns about ₹1.18–1.19 in present value). B has the wider margin of safety on the rate (8.76 vs 6.52 percentage points).

Answer: Both projects are **financially acceptable on their own** and would increase shareholder wealth. If resources allowed, TE Ltd. should do both. Because the projects are mutually exclusive, the decision is about **ranking** them, not about accepting or rejecting them.

Q2. Compare the Rankings

Method	Project A	Project B	Preferred	Margin
NPV at 12%	₹1,86,537	₹1,79,926	**Project A**	A higher by ₹6,611 (3.7%)
IRR	18.52%	20.76%	**Project B**	B higher by 2.24 percentage points
Profitability index	1.1865	1.1799	**Project A**	Equal investment, so it follows NPV

NPV ranks **A first** because it adds more rupee value. IRR ranks **B first** because it earns a higher percentage return. This is a **ranking conflict**: both methods agree that the projects are acceptable but disagree on which is better. NPV compares both projects at the **same rate (12%)**, while IRR compares each project's **own**

break-even rate, so they answer different questions. The conflict is **not caused by size**, since both projects cost ₹10,00,000 and last five years. It is caused by the **timing of the cash flows**.

Q3. Analyse the Conflict

- **Timing.** B is the **front-loaded** project: more of its cash arrives early, which raises its IRR and shortens its payback. A is **back-loaded**: its NPV falls faster as the rate rises (₹28,610 vs ₹20,539 per percentage point), the typical sign of larger cash flows arriving later. Discounted at only 12%, those later flows add up to more value than B's.
- **Crossing NPV profiles.** The profiles cross at about **12.8%**. Below this rate A has the higher NPV, and above it B does. TE Ltd.'s 12% rate is below the crossover, so A creates more value at the rate that actually applies.
- **Reinvestment assumption.** IRR assumes B's early cash can be reinvested at 20.76%. NPV assumes reinvestment at the 12% cost of capital, which is TE Ltd.'s real opportunity rate. The IRR assumption overstates the benefit of getting cash back early.

Answer: B's higher IRR comes from receiving its cash earlier, combined with IRR's assumption that this cash can be reinvested at 20.76%. A generates more value from later cash flows when they are discounted at the actual 12% rate, so its NPV is ₹6,611 higher.

Q4. Evaluate the Decision Criteria

Using NPV, TE Ltd. would select Project A (₹1,86,537 vs ₹1,79,926). **Using IRR, it would select Project B** (20.76% vs 18.52%). The selections differ because:

- **Absolute vs relative measure.** NPV measures value in rupees, while IRR measures a percentage. A higher percentage does not always mean more rupees of value.
- **Different discount rates.** NPV discounts both projects at 12%. IRR in effect discounts each project at its own rate (18.52% and 20.76%), so the comparison is not made on the same basis.
- **Reinvestment and timing.** IRR favours early cash flows because it assumes reinvestment at the IRR. NPV values all cash flows at the 12% cost of capital.
- **Reliability of IRR.** IRR can give several answers (or none) when cash flows change sign more than once. This is not an issue in this case, but it makes IRR less reliable for ranking projects.

Incremental check. Consider the extra cash flows from choosing A instead of B (A minus B). There is no extra outlay, since both cost the same. The NPV of these extra flows at 12% is ₹1,86,537 - ₹1,79,926 = **+₹6,611**, and their IRR equals the crossover rate (about 12.8%), which is above 12%. Applied correctly in this incremental way, **IRR also selects Project A**.

Q5. Recommendation to Management

Recommendation – Select Project A and give priority to NPV. Because the projects are mutually exclusive, TE Ltd. should select **Project A** (NPV ₹1,86,537) and use **NPV as the main decision rule**, treating IRR as supporting information.

- **NPV matches the company's goal.** The aim is to maximise shareholder wealth. A adds ₹6,611 more value in today's money than B. Shareholders benefit from rupees of value, not percentages.
- **NPV uses the right rate.** It discounts the cash flows at TE Ltd.'s actual 12% required return. IRR assumes reinvestment at 20.76%, which the company has no evidence it can achieve.
- **The case conditions favour NPV.** The investment and project life are equal, so the conflict comes only from timing, and NPV handles timing correctly at the company's own cost of capital. The incremental analysis (A minus B) also supports A.
- **Limited capital does not change the answer.** Both projects need the same ₹10,00,000, so the profitability index (A 1.1865 vs B 1.1799) ranks them in the same order: A gives more value per rupee of scarce capital.

Caveats for management. (i) A's lead is small (₹6,611, only 0.66% of the investment), so the cash flow forecasts behind Project A should be checked carefully. (ii) The ranking reverses if the project's true cost of capital is above about 12.8%. If the expansion is riskier than TE Ltd.'s normal business, a higher discount rate would favour B. (iii) A relies more on cash in years 3-5, when demand from automobile and industrial customers is harder to predict, whereas B gets its money back faster. This risk should be handled through a higher discount rate or scenario analysis, not by switching to IRR. A sensitivity analysis, discounted payback and modified IRR (MIRR, which assumes reinvestment at 12%) should support the final approval.

Answer to the Managing Director: "*Choose Project A. When projects are mutually exclusive and NPV and IRR disagree, NPV should decide, because it measures the value added to TE Ltd. at our actual 12% cost of*

capital. Project A adds ₹6,611 more than Project B. B's higher IRR comes mainly from receiving its cash earlier and from assuming that cash could be reinvested at 20.76%."

NPV vs IRR – Comparison of the Two Criteria

Feature	NPV	IRR
What it measures	Absolute value added, in rupees	Break-even rate of return, as a percentage
Discount / reinvestment rate	Company's cost of capital (12%)	The project's own IRR (18.52% / 20.76%)
Link to shareholder wealth	Direct – the increase in firm value	Indirect – a high % can still mean less value
Effect of cash flow timing	Valued correctly at 12%	Favours front-loaded projects (B)
Unusual cash flow patterns	Always gives one answer	Can give several IRRs or none
Additivity	NPVs can be added across projects	IRRs cannot be added
Choice in this case	**Project A**	**Project B** (but the incremental IRR of about 12.8% selects A)

Summary of Answers

Question	Key conclusion
Q1 – Interpretation	Both projects are acceptable on their own: positive NPV, IRR above 12%, profitability index above 1. The real decision is a ranking problem.
Q2 – Rankings	NPV ranks A first (₹1,86,537); IRR ranks B first (20.76%). The conflict comes from timing, not size, since the investments and lives are equal.
Q3 – Cause of conflict	B is front-loaded (higher IRR) and A is back-loaded (higher value at 12%). The NPV profiles cross at about 12.8%, and IRR assumes reinvestment at the IRR.
Q4 – Decision criteria	NPV selects A and IRR selects B. The incremental analysis (A - B) has an NPV of +₹6,611 and an IRR of about 12.8% (above 12%), confirming A.
Q5 – Recommendation	Select **Project A** and rely on **NPV**. Confirm that 12% fits the project's risk and test the year 3-5 forecasts before final approval.